Introduction To Flight Mechanics

Tutorial 11



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Exercise: Basic Controller for the Longitudinal Motion

Given the following state-space representation, design a basic controller for the Boeing 777.

$$\begin{pmatrix} \dot{V}^* \\ \dot{\gamma} \\ \dot{\alpha} \\ \dot{q} \end{pmatrix} = \begin{bmatrix} -0.01 & -0.49 & -0.046 & -0.001 \\ 0.11 & 0.0003 & 1.14 & 0.043 \\ -0.11 & -0.0003 & -1.14 & 0.957 \\ 0.1 & 0 & -15.34 & -3.00 \end{bmatrix} \begin{pmatrix} V^* \\ \gamma \\ \alpha \\ q \end{pmatrix} + \begin{bmatrix} 0.0002 & 0.014 \\ -0.059 & 0 \\ 0.059 & 0 \\ 6.56 & 0.13 \end{bmatrix} \cdot \begin{pmatrix} \eta \\ \delta_T \end{pmatrix}$$

The controller is supposed to augment the short-period eigenmode such that:

- the ideal damping $\zeta_{SP} = \frac{1}{\sqrt{2}}$ and
- the desired eigenfrequency $\omega_{0,SP} = 4{,}5 \frac{rad}{s}$

are achieved.

a) Analyze the eigenmode characteristics of the aircraft without a controller.
b) Propose a suitable controller structure, for which the measurements of the states in the above state space model suffice.
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d) Propose an alternative control structure that accounts for inaccurate measurements of the angle of attack. Compute its required controller gains.

Goal: Given the following state-space representation, **design a basic controller** for the Boeing 777.

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The controller is supposed to augment the short-period eigenmode such that:

- the **ideal damping** $\zeta_{SP} = \frac{1}{\sqrt{2}} \approx 0.7071$ and
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Note:

$$\begin{pmatrix} \dot{V}^* \\ \dot{\gamma} \\ \dot{\alpha} \\ \dot{q} \end{pmatrix} = \begin{bmatrix} -0.01 & -0.49 & -0.046 & -0.001 \\ 0.11 & 0.0003 & 1.14 & 0.043 \\ -0.11 & -0.0003 & -1.14 & 0.957 \\ 0.1 & 0 & -15.34 & -3.00 \end{bmatrix} \begin{pmatrix} V^* \\ \gamma \\ \alpha \\ q \end{pmatrix} + \begin{bmatrix} 0.0002 & 0.014 \\ -0.059 & 0 \\ \mathbf{0.059} & 0 \\ \mathbf{6.56} & 0.13 \end{bmatrix} \cdot \begin{pmatrix} \eta \\ \delta_T \end{pmatrix}$$

The attentive student might notice here that the substitute ***derivatives*** Z_η *and* M_η *have a* ***positive sign****. This does not fit with what we have learned so far, since this would mean that a* ***positive elevator deflection leads to a positive moment****. This is in* ***contradiction*** *to earlier exercises where we learned exactly the opposite.*

The reason for this discrepancy is that when the linear model was generated, the ***control input was not the elevator****, but actually the* ***stick input*** *from the pilot.*

A ***positive stick*** *deflection leads to a* ***negative elevator*** *deflection and thus to a* ***positive moment****. Strictly speaking, the input should therefore not be named* η_c *but "Stick_cmd " instead.*

For our analyses, however, this makes no difference.

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Analyze the **short-period eigenmode** characteristics of the aircraft without a controller.



$$\begin{pmatrix} \dot{V}^* \\ \dot{\gamma} \\ \dot{\alpha} \\ \dot{q} \end{pmatrix} = \begin{bmatrix} -0.01 & -0.49 & -0.046 & -0.001 \\ 0.11 & 0.0003 & 1.14 & 0.043 \\ -0.11 & -0.0003 & -1.14 & 0.957 \\ 0.1 & 0 & -15.34 & -3.00 \end{bmatrix} \begin{pmatrix} V^* \\ \gamma \\ \alpha \\ q \end{pmatrix} + \begin{bmatrix} 0.0002 & 0.014 \\ -0.059 & 0 \\ 0.059 & 0 \\ 6.56 & 0.13 \end{bmatrix} \cdot \begin{pmatrix} \eta \\ \delta_T \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \dot{\alpha} \\ \dot{q} \end{pmatrix} = \begin{bmatrix} Z_\alpha & Z_q + 1 \\ M_\alpha & M_q \end{bmatrix} \cdot \begin{pmatrix} \alpha \\ q \end{pmatrix} + \begin{bmatrix} Z_\eta \\ M_\eta \end{bmatrix} \cdot \eta_c$$

$$Z_\alpha = -1{,}14 \qquad Z_q + 1 = 0{,}957$$

$$M_\alpha = -15{,}34 \qquad M_q = -3{,}00$$

State the characteristic equation and compare it to its general form:

$$\begin{bmatrix} s - Z_\alpha & -(Z_q + 1) \\ -M_\alpha & s - M_q \end{bmatrix}$$

$$\det(s\boldsymbol{I} - \boldsymbol{A}) = s^2 - \underbrace{(M_q + Z_\alpha)}_{2\cdot\sigma = -2\zeta_{SP}\omega_{0,\mathrm{SP}}} \cdot s + \underbrace{M_q \cdot Z_\alpha - M_\alpha \cdot (Z_q + 1)}_{\omega_{0,SP}^2}$$

⇨ Eigenfrequency: $\omega_{0,SP}^2 = Z_\alpha M_q - M_\alpha \cdot (\mathrm{Z_q} + 1)$

⇨ Damping: $\zeta_{SP} = -\frac{1}{2}\frac{Z_\alpha + M_q}{\omega_{0,\mathrm{SP}}}$

$$\omega_{0,SP} = \sqrt{Z_\alpha M_q - M_\alpha \cdot (\mathrm{Z_q} + 1)} = \sqrt{-1{,}14 \cdot (-3{,}00) + 15{,}34 \cdot (0{,}957)} = 4{,}25\,\frac{\mathrm{rad}}{\mathrm{s}}$$

$$\zeta_{SP} = -\frac{1}{2}\frac{(-1{,}14) - 3}{4{,}25} = 0{,}49$$

Desired:

$\zeta_{SP} = \frac{1}{\sqrt{2}} \approx 0.7071$

$\omega_{0,SP} = 4{,}5\,\frac{rad}{s}$

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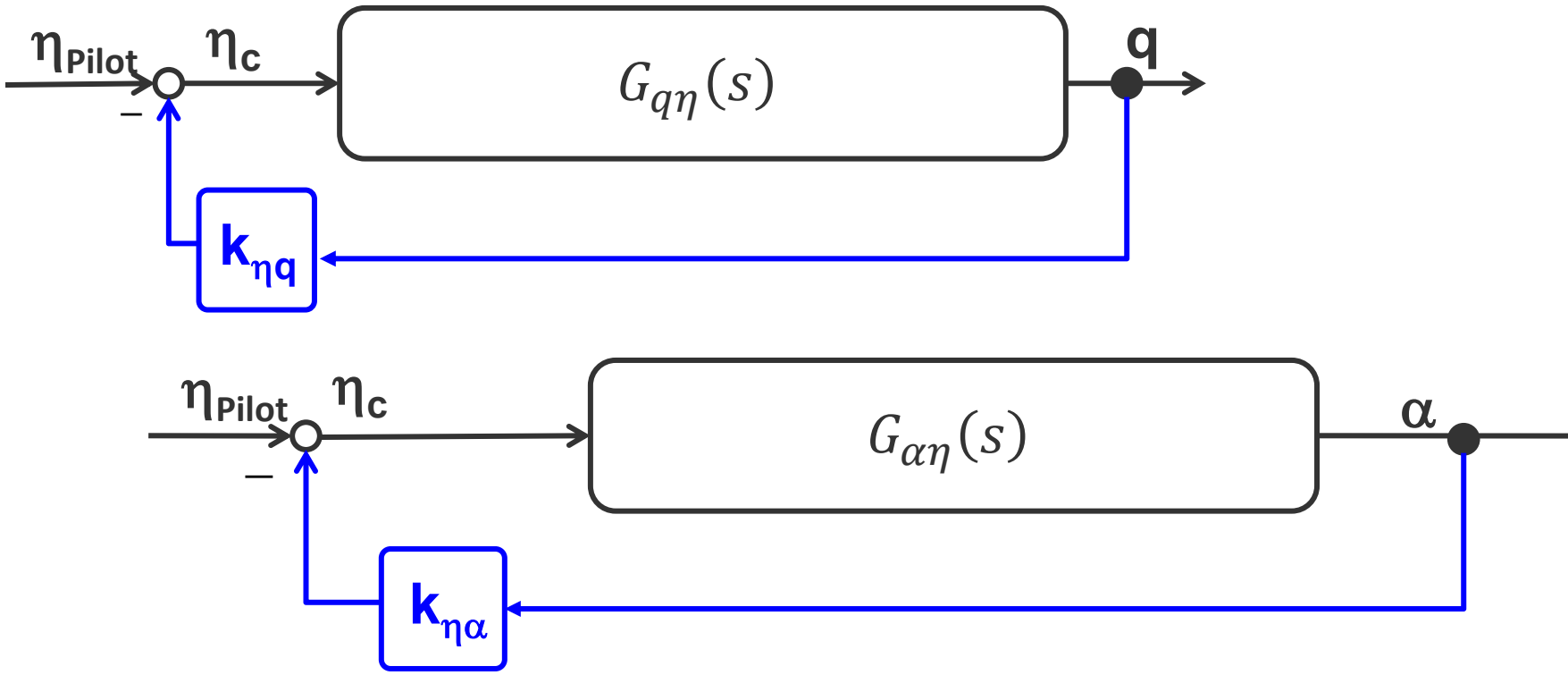
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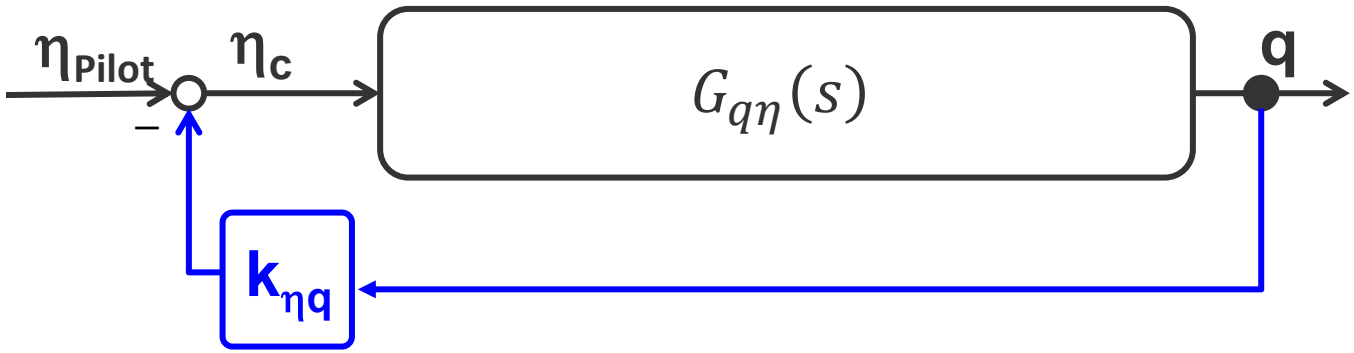
d) Propose an alternative control structure that accounts for inaccurate measurements of the angle of attack. Compute its required controller gains.

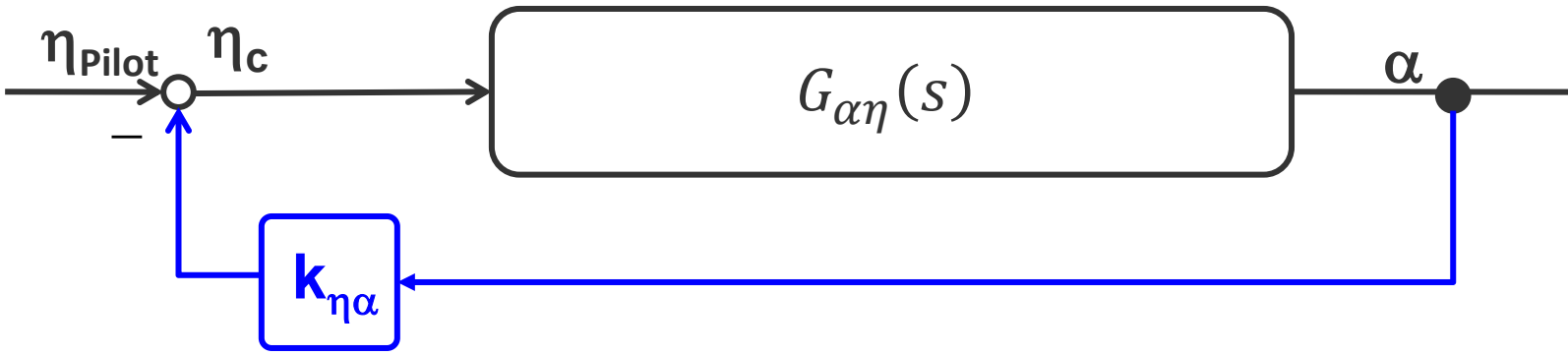
Controller Structure and Root Locus of a Pitch Damper



Feedback	Effect on ω_0	Effect on ζ
pitch rate q		
angle of attack α		

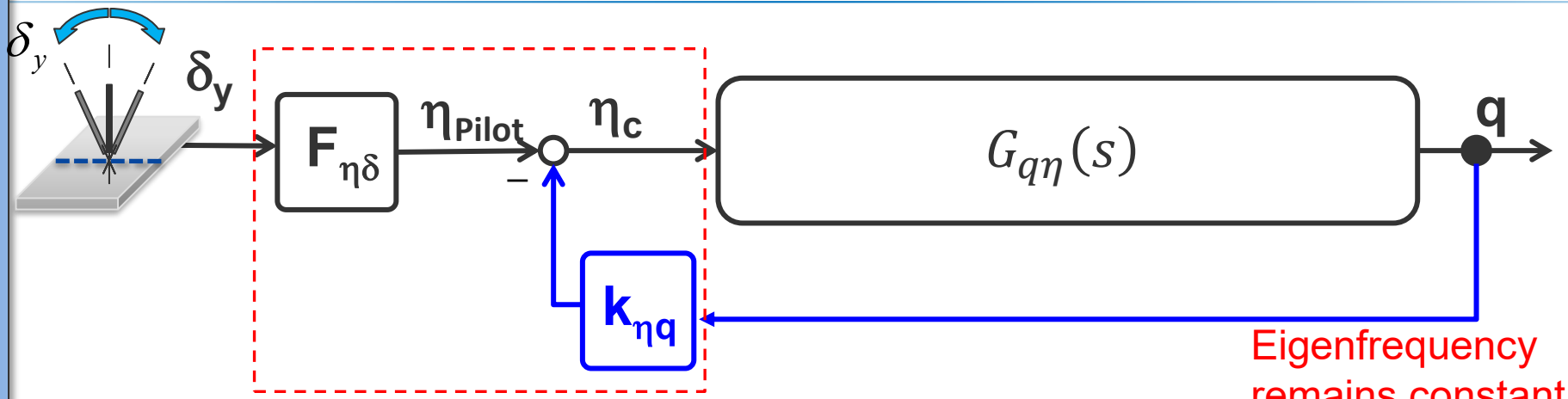
Controller Structure and Root Locus of a Pitch Damper





Feedback	Effect on ω_0	Effect on ζ
pitch rate q	$\approx const$	$increases$
angle of attack α	$increases$	$\approx const$

Controller Structure and Root Locus of a Pitch Damper

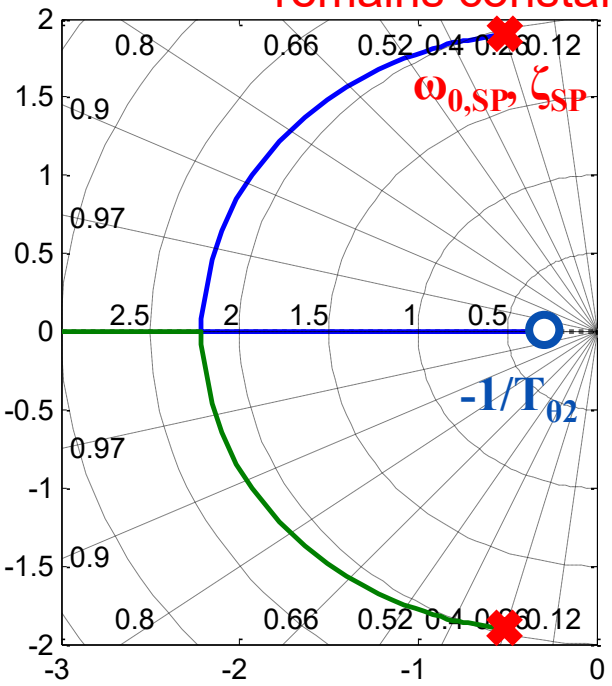


A feedback of the pitch rate to the elevator leads to the following root locus:

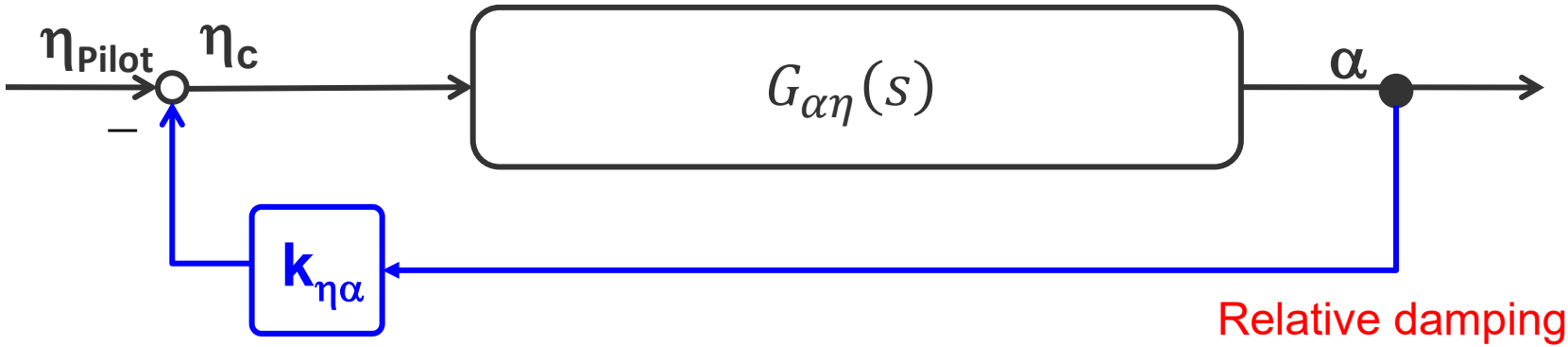
- The natural frequency of the short period remains almost constant.
- Increasing the feedback gain **increases the relative damping of the short period**

Reason: Production of an additional pitch moment proportional to the pitch rate

Eigenfrequency remains constant



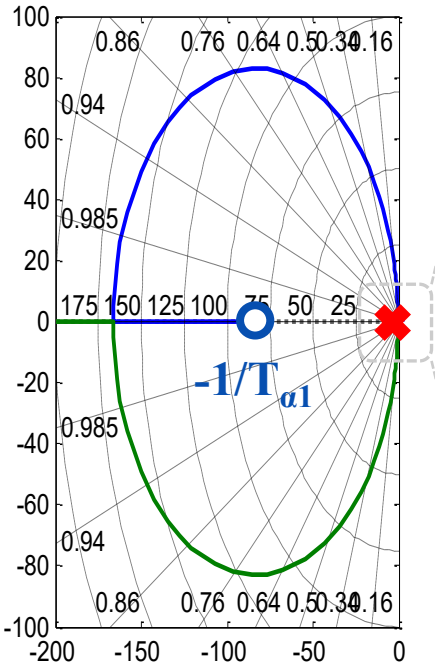
Controller Structure/Root locus α-feedback



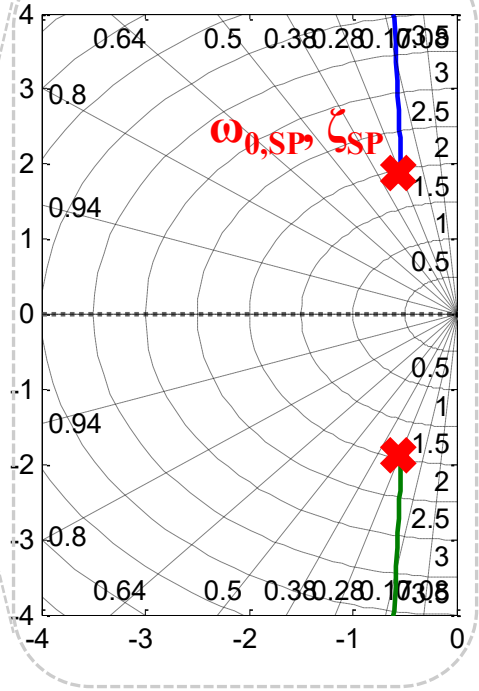
A feedback of the angle of attack to the elevator yields the following root locus:

- For increasing the feedback gain **the eigenfrequency of the short period increases.**
- For reasonable feedback gains, the relative damping remains almost constant.

Reason: Production of an additional moment proportional to the angle of attack

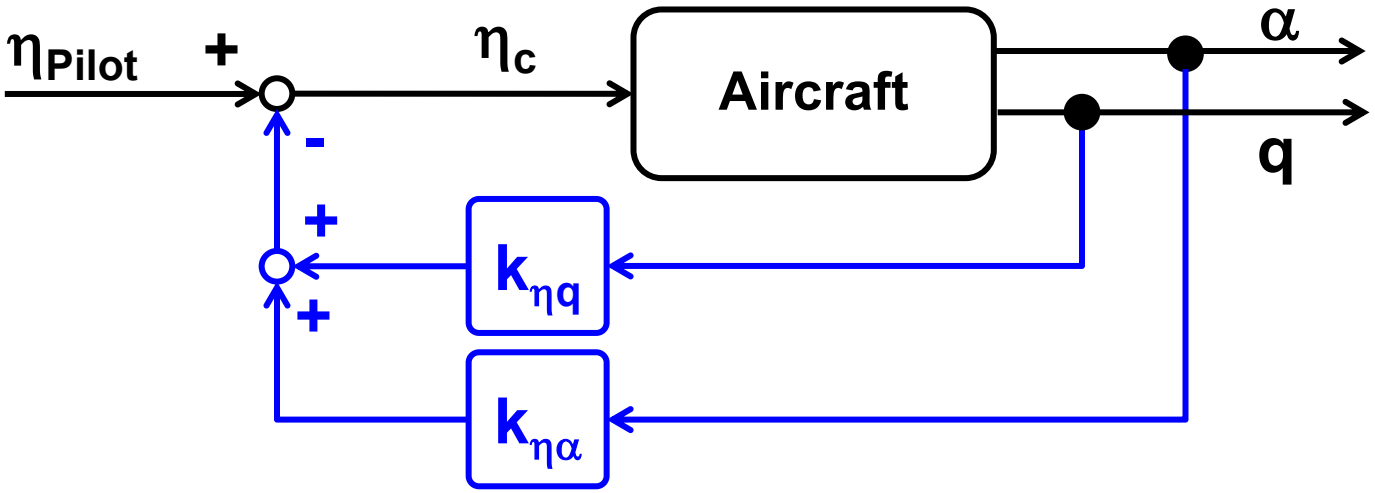


Relative damping remains constant



b) Propose a suitable controller structure.

We require a control structure that has an influence on **damping and eigenfrequency** and uses only the states → combination of a q-feedback and a α-feedback.



Assumption:
q and α are measured sufficiently accurately!

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$$\zeta_{SP,desired} = \frac{1}{\sqrt{2}} \approx 0.7071$$

$$\omega_{0,SP,desired} = 4{,}5\,\frac{rad}{s}$$

Without control augmentation: from a)

$$\det(s\boldsymbol{I}-\boldsymbol{A}) = s^2 - \underbrace{(M_q+Z_\alpha)}_{2\cdot\sigma=-2\zeta_{SP}\omega_{0,SP}}\cdot s + \underbrace{M_q\cdot Z_\alpha - M_\alpha\cdot(Z_q+1)}_{\omega_{0,SP}^2}$$

$$\omega_{0,SP}^2 = Z_\alpha M_q - M_\alpha\cdot(\mathrm{Z_q}+1) = 4{,}25\,\frac{\mathrm{rad}}{\mathrm{s}}$$

$$\zeta_{SP} = -\frac{1}{2}\frac{Z_\alpha+M_q}{\omega_0} = 0{,}49$$

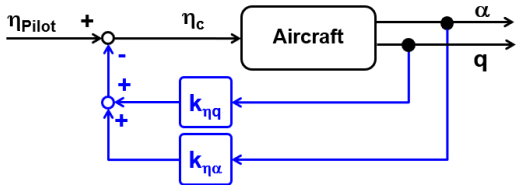
Determine the required controller gains such that the above-mentioned requirements for the short-period are fulfilled.

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & Z_q+1\\ M_\alpha & M_q\end{bmatrix}\cdot\begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}Z_\eta\\ M_\eta\end{bmatrix}\cdot\eta_C$$

$$\zeta_{SP,desired} = \frac{1}{\sqrt{2}} \approx 0.7071$$

$$\omega_{0,SP,desired} = 4{,}5\,\frac{rad}{s}$$

Let‘s assess the **influence of the control law** on the derivatives (i.e. the eigenmotion):



Control Law:

$$\eta_c = \eta_{Pilot} - k_{\eta q}\cdot q - k_{\eta\alpha}\cdot\alpha$$

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & Z_q+1\\ M_\alpha & M_q\end{bmatrix}\cdot\begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}Z_\eta\\ M_\eta\end{bmatrix}\cdot\eta_C$$

How will the closed–loop system matrix look like?

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & Z_q+1\\ M_\alpha & M_q\end{bmatrix}\cdot\begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}Z_\eta\\ M_\eta\end{bmatrix}\cdot\left(\eta_{Pilot} - k_{\eta q}\cdot q - k_{\eta\alpha}\cdot\alpha\right)$$

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & Z_q+1\\ M_\alpha & M_q\end{bmatrix}\cdot\begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}Z_\eta\left(\eta_{Pilot} - k_{\eta q}\cdot q - k_{\eta\alpha}\cdot\alpha\right)\\ M_\eta\left(\eta_{Pilot} - k_{\eta q}\cdot q - k_{\eta\alpha}\cdot\alpha\right)\end{bmatrix}$$

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & Z_q+1\\ M_\alpha & M_q\end{bmatrix}\cdot\begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}Z_\eta\eta_{Pilot} - Z_\eta k_{\eta q}\cdot q - Z_\eta k_{\eta\alpha}\cdot\alpha\\ M_\eta\eta_{Pilot} - M_\eta k_{\eta q}\cdot q - M_\eta k_{\eta\alpha}\cdot\alpha\end{bmatrix}$$

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & Z_q+1\\ M_\alpha & M_q\end{bmatrix}\cdot\begin{bmatrix}\alpha\\ q\end{bmatrix} - \begin{bmatrix}Z_\eta k_{\eta\alpha} & Z_\eta k_{\eta q}\\ M_\eta k_{\eta\alpha} & M_\eta k_{\eta q}\end{bmatrix}\cdot\begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}Z_\eta\\ M_\eta\end{bmatrix}\cdot\eta_{Pilot}$$

$$\mathbf{A}_{CL}$$

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha - Z_\eta k_{\eta\alpha} & Z_q + 1 - Z_\eta k_{\eta q}\\ M_\alpha - M_\eta k_{\eta\alpha} & M_q - M_\eta k_{\eta q}\end{bmatrix} \cdot \begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}Z_\eta\\ M_\eta\end{bmatrix} \cdot \eta_{Pilot}$$

$$\mathbf{A}_{CL} = \begin{bmatrix}Z_\alpha^* & Z_q^* + 1\\ M_\alpha^* & M_q^*\end{bmatrix}$$

$$Z_\alpha^* = Z_\alpha - Z_\eta k_{\eta\alpha} \qquad Z_q^* = Z_q - Z_\eta k_{\eta q}$$

$$M_\alpha^* = M_\alpha - M_\eta k_{\eta\alpha} \qquad M_q^* = M_q - M_\eta k_{\eta q}$$

from a)

$$\omega_{0,SP,desired} = \sqrt{Z_\alpha^* M_q^* - M_\alpha^*(Z_q^* + 1)}$$

$$= \sqrt{(Z_\alpha - Z_\eta k_{\eta\alpha})(M_q - M_\eta k_{\eta q}) - (M_\alpha - M_\eta k_{\eta\alpha}) \cdot (Z_q - Z_\eta k_{\eta\alpha} + 1)}$$

$$\zeta_{SP,desired} = -\frac{1}{2}\frac{Z_\alpha^* + M_q^*}{\omega_{0,SP,desired}} = -\frac{1}{2}\frac{(Z_\alpha - Z_\eta k_{\eta\alpha}) + (M_q - M_\eta k_{\eta q})}{\omega_{0,SP,desired}}$$

2 equations, 2 unknowns (same formulas as always)

Assumption: $Z_\eta \approx 0$

"We make use of this assumption to reduce the computational effort!"



$$\zeta_{SP,desired} = -\frac{1}{2}\frac{\left(Z_\alpha - \cancel{Z_\eta} k_{\eta\alpha}\right) + \left(M_q - M_\eta k_{\eta q}\right)}{\omega_{0,SP,desired}}$$

$$\zeta_{SP,desired} = -\frac{1}{2}\frac{\left(Z_\alpha + M_q - M_\eta k_{\eta q}\right)}{\omega_{0,SP,desired}}$$

$$2 \cdot \zeta_{SP,desired} \cdot \omega_{0,SP,desired} = -Z_\alpha - \left(M_q - M_\eta k_{\eta q}\right)$$

$$k_{\eta q} = \frac{2 \cdot \zeta_{SP,desired} \cdot \omega_{0,SP,desired} + Z_\alpha + M_q}{M_\eta} = 0{,}3389$$

$$\omega_{0,SP,desired}^2 = \left(Z_\alpha - \cancel{Z_\eta} k_{\eta\alpha}\right)\left(M_q - M_\eta k_{\eta q}\right) - \left(M_\alpha - M_\eta k_{\eta\alpha}\right) \cdot \left(Z_q - \cancel{Z_\eta} k_{\eta\alpha} + 1\right)$$

$$\omega_{0,SP,desired}^2 = \left(Z_\alpha\right)\left(M_q - M_\eta k_{\eta q}\right) - \left(M_\alpha\left(Z_q + 1\right) - M_\eta k_{\eta\alpha}\left(Z_q + 1\right)\right)$$

$$\omega_{0,SP,desired}^2 = \left(Z_\alpha\right)\left(M_q - M_\eta k_{\eta q}\right) - M_\alpha\left(Z_q + 1\right) + M_\eta k_{\eta\alpha}\left(Z_q + 1\right)$$

$$k_{\eta\alpha} = \frac{\omega_{0,SP,desired}^2 - Z_\alpha \cdot \left(M_q - M_\eta k_{\eta q}\right) + M_\alpha\left(Z_q + 1\right)}{M_\eta\left(Z_q + 1\right)} = -0{,}0614$$



The controller parameters were calculated under the assumption $Z_\eta = 0$. How could you verify, in a subsequent step, whether this assumption was appropriate for the controller parameter calculation?

How high is the error due to our simplification?
What are the true resulting relative damping and natural frequency values, given our controller parameter values?

$$\omega_{0,SP,full} = \sqrt{(Z_\alpha - Z_\eta k_{\eta\alpha})(M_q - M_\eta k_{\eta q}) - (M_\alpha - M_\eta k_{\eta\alpha}) \cdot (Z_q - Z_\eta k_{\eta\alpha} + 1)} = 4{,}46 \frac{\text{rad}}{\text{s}}$$

Error: $0{,}04 \frac{rad}{s}$ (=0,89%)

Desired:
$\zeta_{SP} = \frac{1}{\sqrt{2}} = 0.7071$
$\omega_{0,SP} = 4{,}5 \frac{rad}{s}$

$$\zeta_{SP,full} = -\frac{1}{2} \frac{(Z_\alpha - Z_\eta k_{\eta\alpha}) + (M_q - M_\eta k_{\eta q})}{\omega_{0,SP,desired}} = 0{,}707$$

Error: 0,0001 (=0,1%),

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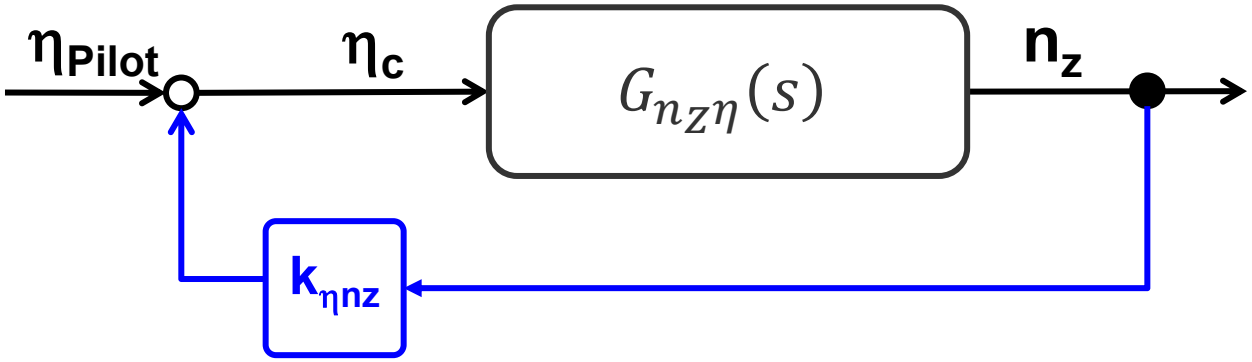
Problem with the quality and availability of measurements?

→ especially for the angle of attack measurement (expensive, accuracy)

2 Alternatives:

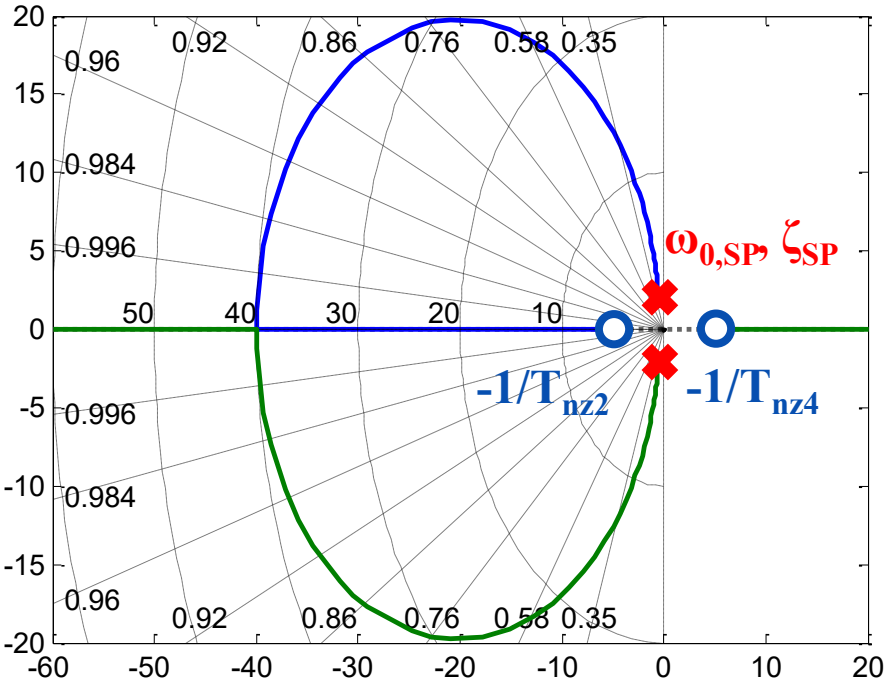
1. Feedback measured load factor (“*load factor is a hidden observation of AoA*”)
2. Filter the pitch rate (pitch controller with lead lag filter)

1) Feedback of Loadfactor



A feedback of the loadfactor leads to the following root locus:

- Increasing the feedback gain of the load factor **increases the natural frequency of the short period.**
- The relative damping remains constant for reasonable feedback gains.



Derivation n_z-feedback

Derive the relation: $\delta n_z = \frac{V_0}{g} \cdot (\delta q - \delta \dot{\alpha})$

Point mass equations of motion in kinematic frame: (Results for a flat, non-rotating earth model)

$$\begin{bmatrix} \left(\dot{V}_K^G\right)^{EK} \\ \left(\dot{\chi}_K^G\right) \cdot \cos\left(\gamma_K^G\right) \cdot \left(V_K^G\right)^E \\ -\left(\dot{\gamma}_K^G\right) \cdot \left(V_K^G\right)^E \end{bmatrix}_K^{EK} = \frac{1}{m} \cdot \sum \begin{bmatrix} X^G \\ Y^G \\ Z^G \end{bmatrix}_K$$

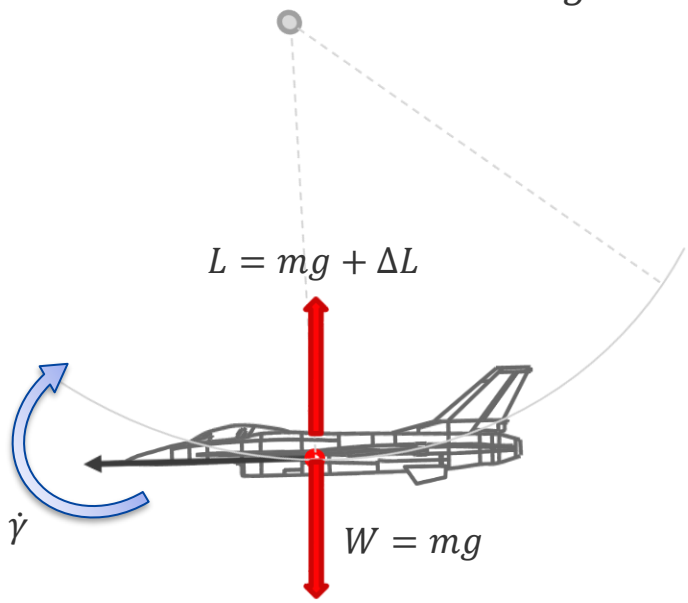
$$\left(\dot{V}_K^G\right)_K^{EK} = \frac{1}{m} \cdot \left(X^G\right)_K$$

$$\left(\dot{\chi}_K^G\right)_K^{EK} = \frac{1}{m \cdot \cos\left(\gamma_K^G\right)_K^E \cdot \left(V_K^G\right)_K^E} \cdot \left(Y^G\right)_K$$

$$\left(\dot{\gamma}_K^G\right)_K^{EK} = -\frac{1}{m \cdot \left(V_K^G\right)_K^E} \cdot \left(Z^G\right)_K$$

$$\dot{\gamma} = -\frac{1}{mV}(-L + mg)$$

$$\delta n_z = n_z - n_{z,0}, \qquad n_z = \frac{L}{mg}$$



Derivation n_z-feedback

Derive the relation: $\delta n_z = \frac{V_0}{g} \cdot (\delta q - \delta \dot{\alpha})$

Point mass equations of motion in kinematic frame: (Results for a flat, non-rotating earth model)

$$\begin{bmatrix} \left(\dot{V}_K^G\right)^{EK} \\ \left(\dot{\chi}_K^G\right) \cdot \cos\left(\gamma_K^G\right) \cdot \left(V_K^G\right)^E \\ -\left(\dot{\gamma}_K^G\right) \cdot \left(V_K^G\right)^E \end{bmatrix}_K^{EK} = \frac{1}{m} \cdot \sum \begin{bmatrix} X^G \\ Y^G \\ Z^G \end{bmatrix}_K$$

$$\left(\dot{V}_K^G\right)_K^{EK} = \frac{1}{m} \cdot \left(X^G\right)_K$$

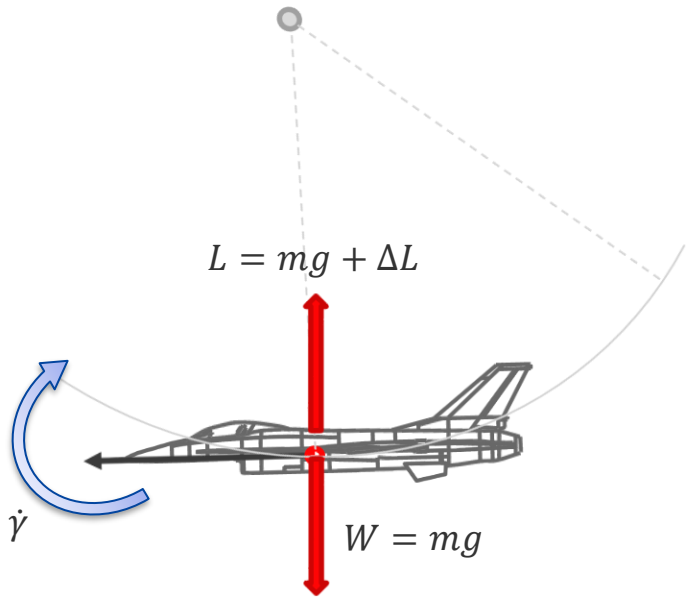
$$\left(\dot{\chi}_K^G\right)_K^{EK} = \frac{1}{m \cdot \cos\left(\gamma_K^G\right)_K^E \cdot \left(V_K^G\right)_K^E} \cdot \left(Y^G\right)_K$$

$$\left(\dot{\gamma}_K^G\right)_K^{EK} = -\frac{1}{m \cdot \left(V_K^G\right)_K^E} \cdot \left(Z^G\right)_K$$

$$\dot{\gamma} = -\frac{1}{mV}(-L + mg)$$

$$\Rightarrow \quad \dot{\gamma} = \frac{g}{V}\left(\frac{L-mg}{mg}\right) = \frac{g}{V}\left(\frac{L-L_0}{mg}\right) = \frac{g}{V}\left(n_z - n_{z,0}\right) = \frac{g}{V}\delta n_z$$

$$\Rightarrow \quad \delta n_z = \frac{V}{g}\dot{\gamma} = \frac{V}{g}\left(\dot{\Theta} - \dot{\alpha}\right) = \frac{V}{g}(q - \dot{\alpha})$$



This applies only to small angles and deviations from level flight.

Derivation n_z-feedback

$$\delta n_z = \frac{V_0}{g} \cdot (\delta q - \delta\dot{\alpha}) \xrightarrow[Z_q = 0]{\delta\dot{\alpha} = Z_\alpha \cdot \delta\alpha + \delta q + Z_\eta \delta\eta_C} \delta n_z = \frac{V_0}{g} \cdot \left(\delta q - \left(Z_\alpha \cdot \delta\alpha + \delta q + Z_\eta \delta\eta_C\right)\right)$$

$$\boxed{\delta n_z = -\frac{V_0}{g} \cdot \left(Z_\alpha \cdot \delta\alpha + Z_\eta \delta\eta_C\right)} \longleftrightarrow \boxed{\delta\alpha = \frac{-\frac{g}{V_0} \cdot \delta n_z + Z_\eta \delta\eta_C}{Z_\alpha}}$$

With this step we have a simplified relation between $\delta\alpha$ und δn_z (under the assumption $Z_q = 0$).

$$\delta\dot{n}_z = -\frac{V_0}{g} \cdot \left(Z_\alpha \cdot \delta\dot{\alpha} + Z_\eta \delta\dot{\eta}_C\right)$$

$$\downarrow \quad \delta\dot{\alpha} = Z_\alpha \cdot \delta\alpha + \delta q + Z_\eta \delta\eta_C,$$

$$\delta\dot{n}_z = -\frac{V_0}{g} \cdot \left(Z_\alpha \cdot \left(Z_\alpha \cdot \delta\alpha + \delta q + Z_\eta \delta\eta_C\right) + Z_\eta \delta\dot{\eta}_C\right)$$

$$\downarrow \quad \delta n_z = -\frac{V_0}{g} \cdot \left(Z_\alpha \cdot \delta\alpha + Z_\eta \delta\eta_C\right)$$

$$\delta\dot{n}_z = Z_\alpha \delta n_z - \frac{V_0}{g} Z_\alpha \delta q - \frac{V_0}{g} Z_\eta \delta\dot{\eta}_C$$

This $\delta\dot{n}_z$-equation with the states δn_z und δq and the time derivative $\delta\dot{\eta}_C$ replaces the $\delta\dot{\alpha}$-equation.

Derivation n_z-feedback

If we assume additionally $Z_\eta \approx 0$ then we can formulate the state space model as follows:

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & Z_q + 1\\ M_\alpha & M_q\end{bmatrix} \cdot \begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}0\\ M_\eta\end{bmatrix} \cdot \eta_C$$

$$\delta\dot{n}_z = \mathrm{Z}_\alpha \delta n_z - \frac{V_0}{g}\mathrm{Z}_\alpha \delta q - \frac{V_0}{g} Z_\eta \delta\dot{\eta}_C$$

$$Z_\eta \approx 0$$

$$\begin{bmatrix}\delta\dot{n}_z\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & -\frac{V_0}{g}\mathrm{Z}_\alpha\\ ? & M_q\end{bmatrix} \cdot \begin{bmatrix}\delta n_z\\ q\end{bmatrix} + \begin{bmatrix}0\\ M_\eta\end{bmatrix} \cdot \eta_C$$

$$\delta\alpha = \frac{\frac{g}{V_0}\cdot\delta n_z - Z_\eta \delta\eta_C}{\mathrm{Z}_\alpha} \overset{Z_\eta \approx 0}{=} \frac{g}{V_0 Z_\alpha}\cdot\delta n_z$$

$$M_\alpha \cdot \alpha = M_\alpha \frac{g}{V_0 Z_\alpha}\cdot\delta n_z$$

Relation from previous slide

$$\begin{bmatrix}\delta\dot{n}_z\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & -\frac{V_0}{g}\mathrm{Z}_\alpha\\ M_\alpha \frac{g}{V_0 Z_\alpha} & M_q\end{bmatrix} \cdot \begin{bmatrix}\delta n_z\\ q\end{bmatrix} + \begin{bmatrix}0\\ M_\eta\end{bmatrix} \cdot \eta_C$$

Derivation n_z-feedback

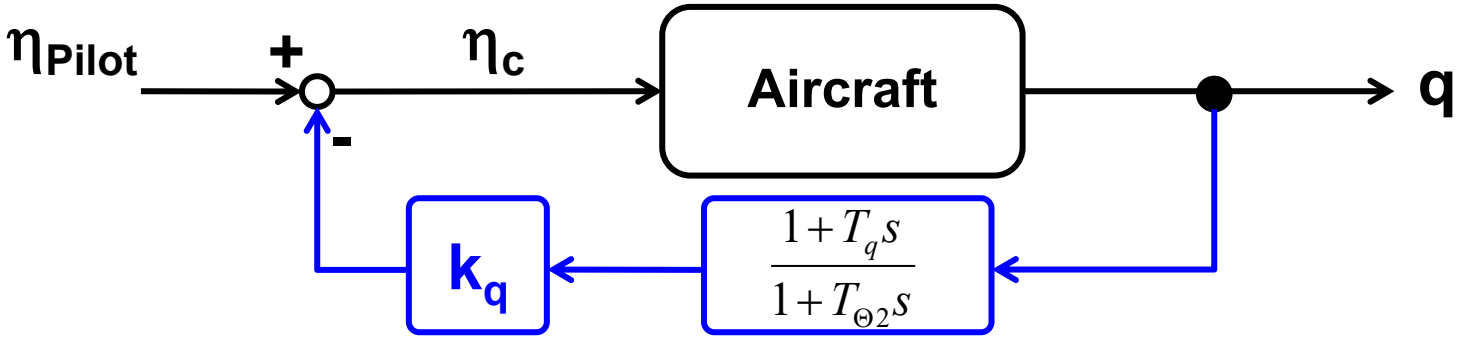
$$\begin{bmatrix} \delta\dot{n}_z \\ \dot{q} \end{bmatrix} = \begin{bmatrix} Z_\alpha & -\frac{V_0}{g} Z_\alpha \\ M_\alpha \frac{g}{V_0 Z_\alpha} & M_q \end{bmatrix} \cdot \begin{bmatrix} \delta n_z \\ q \end{bmatrix} + \begin{bmatrix} 0 \\ M_\eta \end{bmatrix} \cdot \eta_C$$

$$\eta_C = \eta_{Pilot} - k_{\eta q} \cdot q - k_{\eta n_z} \cdot \delta n_z$$

Proceed as was presented for controller with angle of attack feedback…

However, exercise does not mention that we are allowed to use additional measurements. According to b) only measurements of the states are available. Hnece, we will focus on the variant with pitch feed back and lead-lag filter

2) Pitch controller with lead lag filter in the feedback



Lead lag filter to estimate α from q:

(see recap on next slide)

$$\delta\alpha = \frac{T_{\Theta 2}}{(T_{\Theta 2}s + 1)} \cdot \delta q$$

$$T_{\Theta 2} \approx -\frac{1}{Z_\alpha}$$

Derivation Pitch Controller

Another feedback possibility can be obtained by looking at the relation between angle of attack and pitch rate from the 2x2 approximation:

$$\dot{\alpha} = Z_\alpha \alpha + q \qquad \text{with} \qquad Z_q = Z_\eta = \dot{\gamma}_0 = \gamma_0 = 0$$



Derive the relation given below

Laplace Transformation:

$$s\alpha = Z_\alpha \alpha + q \longrightarrow (s - Z_\alpha)\alpha = q$$

Transfer function $G_{\alpha q}(s)$:

$$G_{\alpha q}(s) = \frac{\alpha(s)}{q(s)} = \frac{1}{s - Z_\alpha} = \frac{-1/Z_\alpha}{-1/Z_\alpha \cdot s + 1} = \frac{T_{\Theta_2}}{T_{\Theta_2} s + 1}$$

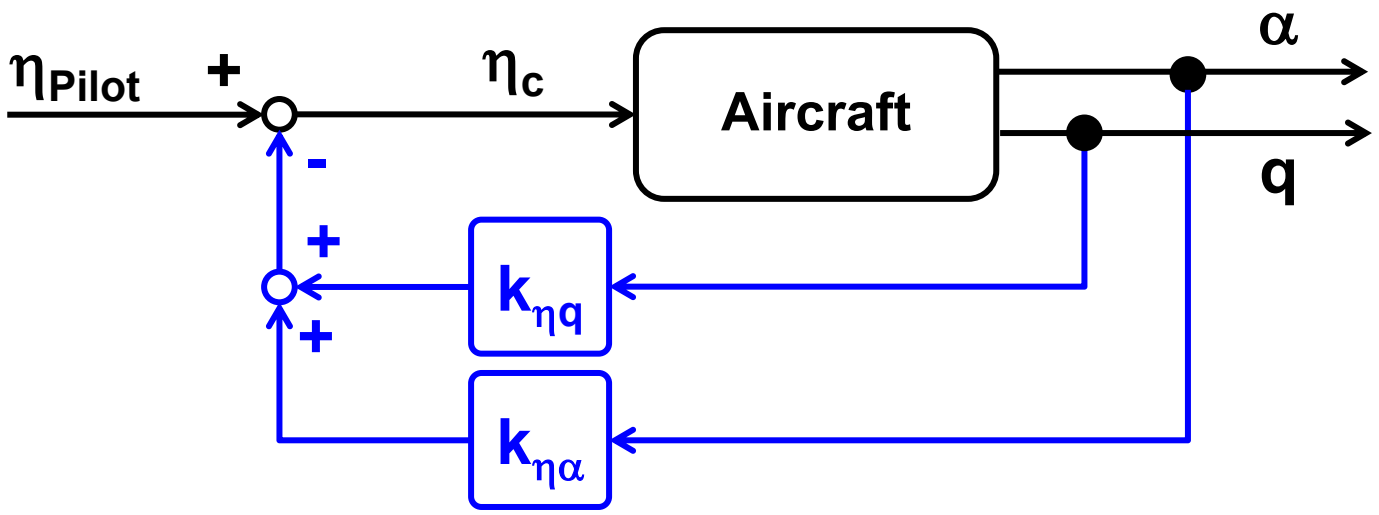
$$\alpha(s) = \frac{T_{\Theta_2}}{T_{\Theta_2} s + 1} q(s)$$

$$T_{\Theta_2} = -1/Z_\alpha$$

Reminder: The assumptions for these equations are:

1. Steady state cruise: $\dot{\gamma}_0 = 0, \gamma_0 = 0$
2. $Z_\eta = 0$
3. $Z_q + 1 = 1$

Controller Structure Pitch Controller



Relation between AoA and q

$$\delta\alpha = \frac{T_{\Theta 2}}{1 + T_{\Theta 2} s} \cdot \delta q$$

with

$$T_{\Theta 2} \approx -\frac{1}{Z_\alpha}$$

Integrate lead lag filter in control law and rearrange:

$$\eta_c = \eta_{Pilot} - \left(k_{\eta q} \cdot \delta q + k_{\eta \alpha} \cdot \delta\alpha\right)$$
$$= \eta_{Pilot} - \left(k_{\eta q} \cdot \delta q + k_{\eta \alpha} \cdot \frac{T_{\Theta 2}}{(T_{\Theta 2} s + 1)} \cdot \delta q\right)$$

$$\eta_c = \eta_{Pilot} - \left(k_{\eta q} + k_{\eta \alpha} \cdot \frac{T_{\Theta 2}}{(T_{\Theta 2} s + 1)}\right) \cdot \delta q$$

$$\eta_c = \eta_{Pilot} - \left(k_{\eta q} + k_{\eta \alpha} \cdot \frac{T_{\Theta 2}}{(T_{\Theta 2} s + 1)} \right) \cdot \delta q$$

$$\eta_c = \eta_{Pilot} - \left(\frac{k_{\eta q}(T_{\Theta 2} s + 1)}{(T_{\Theta 2} s + 1)} + \frac{k_{\eta \alpha} \cdot T_{\Theta 2}}{(T_{\Theta 2} s + 1)} \right) \cdot \delta q$$

$$\eta_c = \eta_{Pilot} - \left(\frac{k_{\eta q}(T_{\Theta 2} s + 1) + k_{\eta \alpha} \cdot T_{\Theta 2}}{(T_{\Theta 2} s + 1)} \right) \cdot \delta q$$

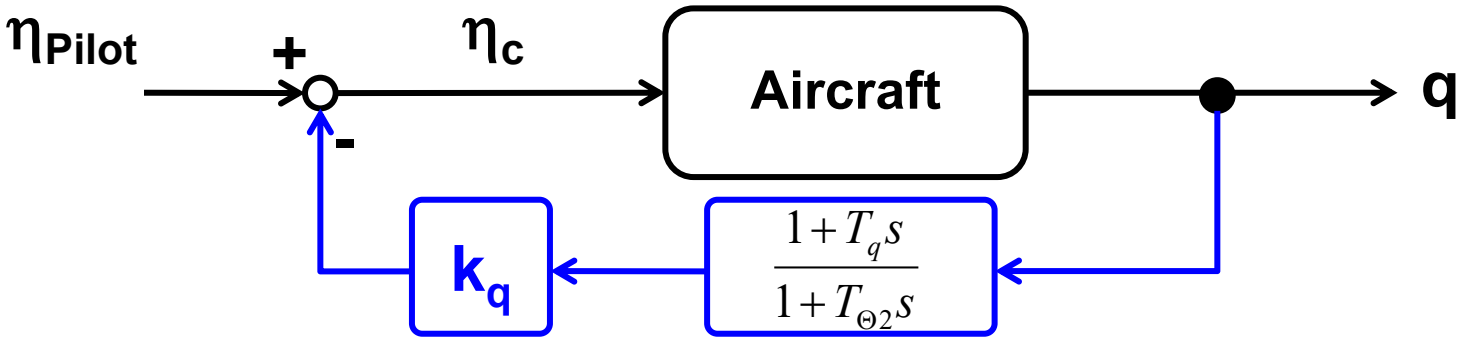
$$\eta_c = \eta_{Pilot} - \left(\frac{k_{\eta q} T_{\Theta 2} s + k_{\eta q} + k_{\eta \alpha} \cdot T_{\Theta 2}}{(T_{\Theta 2} s + 1)} \right) \cdot \delta q$$

$$\eta_c = \eta_{Pilot} - \left(\underbrace{(k_{\eta q} + k_{\eta \alpha} \cdot T_{\Theta 2})}_{k_q} \frac{\left(\underbrace{\frac{k_{\eta q} T_{\Theta 2}}{k_{\eta q} + k_{\eta \alpha} \cdot T_{\Theta 2}}}_{T_q} s + 1 \right)}{(T_{\Theta 2} s + 1)} \right) \cdot \delta q$$

$$\eta_c = \eta_{Pilot} - \left(\underbrace{\left(k_{\eta q} + k_{\eta \alpha} \cdot T_{\Theta 2}\right)}_{k_q} \frac{\left(\underbrace{\frac{k_{\eta q} T_{\Theta 2}}{k_{\eta q} + k_{\eta \alpha} \cdot T_{\Theta 2}}}_{T_q} s + 1 \right)}{(1 + T_{\Theta 2} s)} \right) \cdot \delta q$$

We can now represent the controller using a gain and a filter

Draw the block diagram



We use the gains computed earlier

$$T_{\Theta_2} = -1/Z_\alpha$$

$$k_q = k_{\eta q} + k_{\eta\alpha} \cdot T_{\Theta 2} = 0{,}3389 + \left(-0{,}0613 \cdot \frac{1}{1{,}14}\right) = 0{,}2851$$

$$T_q = \frac{k_{\eta q} T_{\Theta 2}}{k_{\eta q} + k_{\eta\alpha} \cdot T_{\Theta 2}} = \frac{0{,}3389 \frac{1}{-1{,}14}}{0{,}3389 + \left(-0{,}0613 \cdot \frac{1}{+1{,}14}\right)} = 1{,}043$$

